When Models Train on Their Own Homework : Dose-Response of Model Collapse in a Word-Level Transformer and the 75% Human-Text Line

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Abstract

When language models are trained on text written by earlier models, quality, diversity and factual reliability can decline across generations. This is called model collapse. This paper quantifies how quickly the erosion occurs in a small system, and identifies how much human written text each generation's training data needs to keep the vocabulary and diversity close to the original.

For this, A word-level nanoGPT-style transformer was trained from scratch on CPU for three generations. The training data was 13,920 tokens from the opening passages of Charles Dickens's 1859 novel *A Tale of Two Cities*, and the result was evaluated against the validation data of 1,579 tokens from the following passages. Two arms were then trained recursively for the three generations. In the collapse arm, each generation trains only on the previous generation's output. In the mitigated arm, each generation mixes fresh real text with a synthetic share s of that output. Two experiments cover nine values of s, five seeds each, 45 runs in all, and every result below is a mean with a 95% confidence interval.

The collapse arm loses two-thirds of its rare words by generation 3 (retention fell from 0.652 to 0.208), and the vocabulary of generated text shrank by 59.6% (1,821 to 736 words). However, mixing in real data reduced the loss. At synthetic shares of 0.1 to 0.25, models kept 72-88% of their original rare words coverage, and no run produced a repetition loop. At $s \geq 0.3$, 6 of 20 runs produced repetition loops. Vocabulary per set fell with synthetic share. Average metrics barely separated $s = 0.25$ from $s = 0.3$, but worst-case sets did, which is exactly why worst-case checking belongs in every collapse evaluation.

Real data protects against collapse in proportion to its share. For this setup, keeping at least 75% human-written data per generation kept the model near the real-text floor. The thresholds should be read as indicative: one novel, one small model, five seeds.

Keywords

model collapse, nanoGPT, synthetic data, Validation text, recursive training, language models, worst-case metrics, repetition loops

1. Introduction

Nearly every large language model in service today learned its language from text sitting on the public web, and for most of that history almost all of those words were typed by people. That assumption is quietly coming apart. Today, a growing share of content online (web pages, blog posts, news summaries) is written by language models, and that output is exactly what the next round of training will consume. The loop is already in place: machines reading machines, with the human corpus diluted a little more at every pass.

This paper studies the recursive training loop in its smallest honest form (Figure 1). A model never reproduces its input perfectly. It loses some rare words, prefers common phrasing and makes some small sampling errors. Train the next generation on that copy and it inherits the errors plus new ones. Do that for a few generations and the errors compound, and the output drifts further and further from the original distribution. Shumailov and colleagues named the effect model collapse and showed that feeding model-generated content into training leaves irreversible defects, specifically the disappearance of the tails of the original content distribution [1].

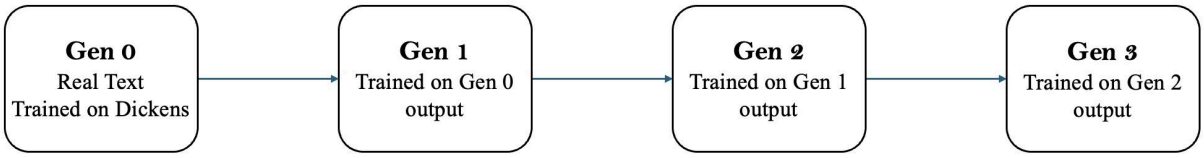


Figure 1: The recursive training loop studied here. Generation n + 1 is trained from scratch on the 13,920 tokens that generation n produced.

1.1 Background

The literature that makes this study possible arrived in a short window and points in consistent directions. Shumailov, I. et al. demonstrated collapse in language models and Gaussian mixtures, and showed that it arises from statistical, functional and approximation errors compounding over generations [1]. Alemohammad et al. observed the same thing in image generators and named such loops self-consuming, showing that once the supply of fresh real data runs short, either the quality or the diversity of the output degrades at every round [2]. The theory came from two sides. Bertrand et al. proved the stability conditions for iterative retraining: the starting model must approximate the data well, and each round must keep a large enough fraction of real data [3]. Gerstgrasser et al. showed empirically and analytically that the failure appears when synthetic output replaces the human text, but goes away when each generation's synthetic output is kept in the mix alongside the original [4]. Dohmatob et al. connected the collapse to the thinning of the distribution's tail and argued that the effect rewrites the scaling laws tying performance to data size [5].

1.2 Research Gap

Each of those papers confirms the same two things: that the failure mode is real, and that genuine data in the mix can stop it. What none of them gives a practitioner is a number: how large the real share has to be, generation by generation, in an actual training pipeline. The existing work reports effects at the extremes, all real data versus all synthetic data, or stability thresholds derived from theory, but a measured dose-response curve, the protection you get as a function of the synthetic share, is missing. There is a second gap in how the failure is measured. Most studies report means across samples, and rarely look for whether those means are hiding outright breakdowns, a generated sample that degenerates into a repetition loop while the average still looks healthy.

1.3 Research Question

Two questions drive the study :

1. What minimum share of each generation's training data must remain human-written to keep rare vocabulary and diversity close to the original?
2. Do mean diversity metrics and worst-case metrics identify the same safe limit?

1.4 Objectives

Four objectives structure the work :

1. To reproduce model collapse in a small model by training it on text generated by the previous generation for three generations.
2. To measure the effect of the synthetic share from 0 to 90% on the collapse.
3. To compare mean and worst-case metrics and identify breakdowns, like repetition loops
4. To find the smallest share of human-written data that keeps the model near its original distribution.

2. Hypotheses

H1 - Collapse : Training on the previous generation's output only (collapse arm) reduces rare-word retention and distinct vocabulary across generations.

H2 - Dose-Response : Vocabulary and diversity decline as the synthetic share s rises, and the decline is non-linear, with a threshold between s= 0.25 and s= 0.5.

H3 - Worst Case : Mean metrics do not reveal the failure threshold; worst-case sets (repetition loops) locate it more precisely.

3. Methodology

3.1 Data

The corpus is the 1859 novel *A Tale of Two Cities* by Charles Dickens, freely distributed by Project Gutenberg [6]. The opening chapters of the novel, starting with the chapter "The Period" and its famous first line, form the training text. The validation text is the stretch of novel immediately after, picked up

mid-sentence at "…hand, and mutter a tremendous curse." The split keeps the evaluation honest: the model has to write text that still reads like the same book even though it never trained on those passages.

One novel does double duty here. It carries a long tail of rare words, so the loss of rare material can be measured word by word. And it is small enough that every model trains in minutes on a CPU, and every output is short enough to read by eye and flag loops manually as well as by metric.

Both the training text and the validation text were lowercased, and split into words and punctuation using the expression $[a - zA - Z'] + |[. , ! ? ; :]$. The vocabulary is built from the training set only and any unseen words are mapped to an <unk> token.

3.2 Model

The model follows the nanoGPT recipe, the minimal word-level design in Karpathy's repository [8], implemented in PyTorch on top of the transformer architecture of Vaswani and colleagues [7]. Every generation is trained from scratch on a CPU, so what one generation inherits from the last is data only, never weights. That is the point of the setup: the question is what recursion to data does, and fresh weights keep the comparison clean. Every model generated 13,920 tokens, the same volume that generation 0 was trained on, so the dose of data is constant across the study.

The architecture, fixed in all runs: three transformer layers, four attention heads, an embedding of width 96, a context window of 48 tokens, 0.1 dropout, AdamW at a learning rate of 0.003, batches of 32, and 1,200 training steps. Sampling at generation time ran at temperature 0.9 with top-k 40. 40.

3.3 Design

Generation 0 is trained completely on real text and is shared by both arms (Figure 2). Every later generation 1 to 3 is trained on:

- **Collapse arm:** entirely on the previous generation's 13920-token output.
- **Mitigated arm:** (1 − s) x 13920-token real text plus s x 13920-token synthetic text (s = synthetic share). The synthetic tokens are from the start of the previous mitigated arm's output.

At s = 0, the mitigated arm is trained on the entire real text every generation, hence, it measures the noise floor. The collapse arm corresponds to s = 1.

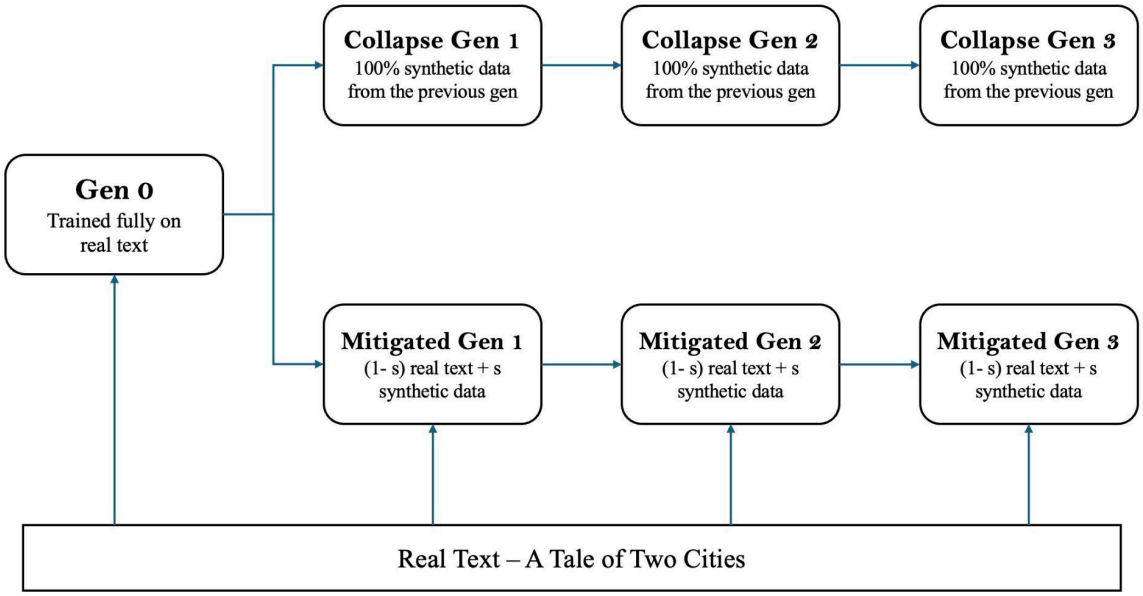


Figure 2 : Experimental design. Generation 0 is trained only on real text and is shared by both arms. In the collapse arm (top), each generation is trained entirely on the previous generation's output. In the mitigated arm (bottom), each generation is trained on synthetic share s from the previous mitigated generation's output.

3.4 Testing

The synthetic share was tested in two stages. Experiment 1 tested $s \in \{0, 0.1, 0.3, 0.5, 0.7, 0.9\}$ where s = 0.1 showed no breakdown and s = 0.3 produced a loop by generation 3. This placed the safe limit between 90% and 70% of real text. So experiment 2 was conducted to test $s \in \{0.15, 0.2, 0.25\}$ and to locate the limit more precisely. Both stages ran the same code on the same data with the same five seeds, 1337 through 1341. That means the generation-0 model and the collapse arm are bit-for-bit identical in every condition; the share s is the only thing that moves.

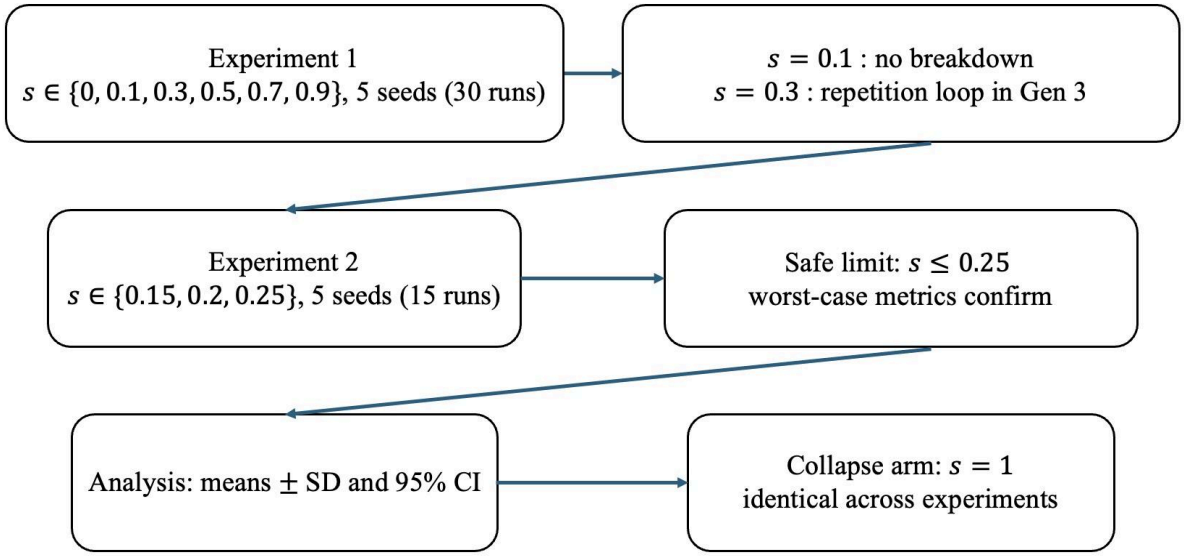


Figure 3 : Experimental flow. The first experiment located a region between s = 0.1 and s = 0.3; the second refined it. Total: 45 runs.

3.5 Metrics

Each output of 13,920 tokens was split into eight sets of 1579 tokens to match the size of the validation text. All metrics were measured for each set, and then the mean over all the sets and the measures of the worst set for every metric were stored.

Having the same length during comparison helps to eliminate any bias that text length might bring to the word counts and distinct-n ratios.

- Distinct Vocabulary: The number of distinct words in each set
- Distinct-n : The share of n-grams in a set that are unique (for n = 2 and n = 3)

$$Distinct - n = \frac{\{unique\ n{-}grams\}}{\{all\ n{-}grams\}}$$

- Rare-Word Retention: The share of 2326 rare words from the training text (set R) that appear in the generated 13920-token text (set S)

$$RR = \frac{R \cap S}{R}$$

- Jensen-Shannon Divergence: The divergence between the word-frequency distribution P of a set and the word-frequency distribution Q of the validation text, in bits, where M = (P + Q)/2. It ranges from 0 (identical) to 1 (no overlap) (JS = Jensen-Shannon Divergence, KL = Kullback–Leibler Divergence)

$$JS(P||Q) = \frac{1}{2} KL(P||M) + \frac{1}{2} KL(Q||M)$$

- Validation Perplexity: the exponential of the mean cross-entropy over 40 random batches of validation sets, excluding <unk> targets
- Repetition Loop: a set with 50 ≥ distinct words

3.6 Analysis

Every number below is a mean over the five seeds with its spread, and a run is declared failed at share s if any window, in any generation, looped. That definition is what lets the worst-case question be asked at all: a condition may average out healthy while still containing a replicate that has clearly broken .

4. Results

4.1 Model Collapse is Reproduced (H1)

The collapse arm declines on every axis, generation after generation (Table 1). Rare-word retention starts at 0.652 for generation 0 and is down to 0.208 by generation 3, so two-thirds of the rare tail is gone. The whole vocabulary contracts from 1,821 distinct words to 736, a 59.6% loss. Bigrams and trigrams follow the same curve: distinct-2 loses 35% and distinct-3 loses 35.8%, consistent with the theoretical account that the tails of the distribution are what collapse first [1][5]. The control tells the other half of the story. Retrained for three generations on 100% real text, the model still holds 1,819 distinct words against the original 1,821, and every other metric sits at nearly its starting value. The machinery of retraining is not what is doing the damage; recursion to the model's own output is. Three generations of that, and two-thirds of the rare words are gone.

Condition	Distinct Vocabulary (Entire)	Vocabulary per set (Mean)	Vocabulary per set (Min)	Distinct 2 (Mean)	Distinct 3 (Mean)	JS Divergence (Mean)	Validation Perplexity (Mean)
Gen 0 (Reference Data)	1821	550.2	513.0	0.772	0.911	0.334	32745
Collapse (Gen 1)	1312	499.2	428.2	0.687	0.814	0.343	46,988
Collapse (Gen 2)	1012	437.9	339.8	0.591	0.694	0.353	60,585
Collapse (Gen 3)	736	381.0	273.4	0.502	0.584	0.361	53757
100% Real Text (Gen 3)	1819	547.1	499.6	0.769	0.904	0.338	34996

Table 1 : Collapse arm (synthetic share s = 1): metrics in the generated text by generation, against the 100%-real control at generation 3.

4.2 Effect of Synthetic Share (H2)

During model collapse, rare words are the first to be lost. However, the more real text each generation kept, the less it lost. But at s = 0.9, it is indistinguishable from pure recursion.

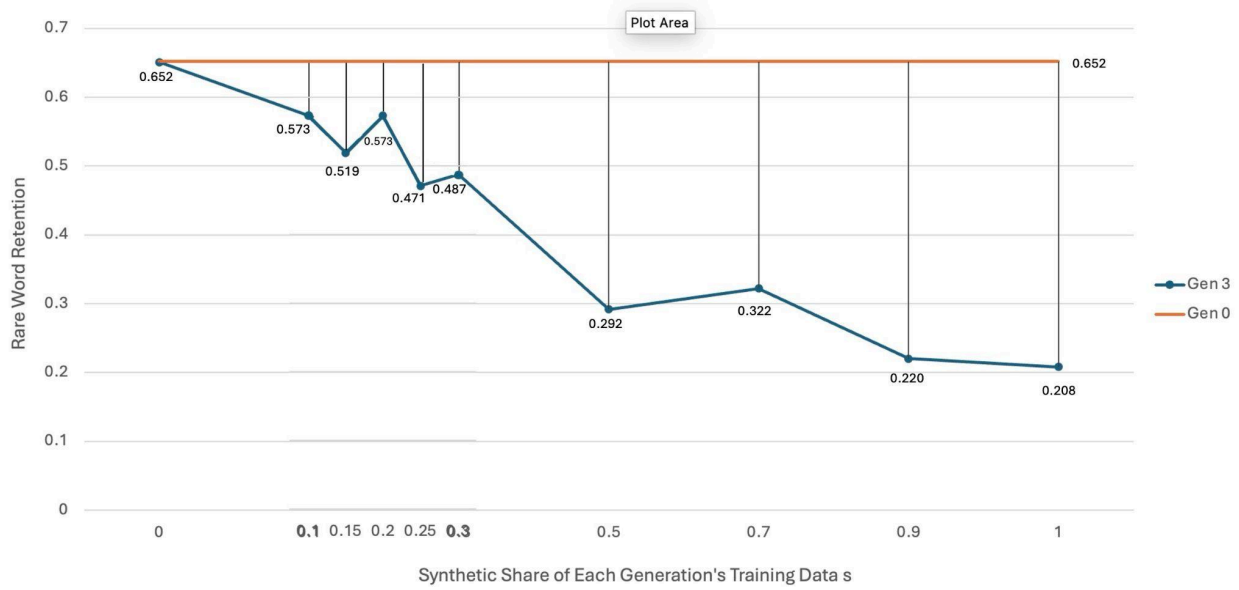


Figure 4 : Rare word retention per 1,579-token set at generation 3, against synthetic share s, with 95% confidence intervals (five seeds).

At generation 3, all other diversity metrics also rank the conditions in nearly the same order: more real text, less collapse. At s = 0.1 to 0.3, average diversity drops by only a little (distinct-2 fell from 0.77 to 0.65–0.73, rare-word retention fell to 72 - 88%). However, at s ≥ 0.5, the runs were close to the collapse arm (distinct-2 fell from 0.77 to 0.46 - 0.49, rare-word retention fell to 34% - 49%).

The small reversal between neighbouring synthetic shares, s = 0.15 and 0.2 and s = 0.5 and 0.7, is within one standard deviation, so ordering within those pairs should not be interpreted.

Condition	Vocabulary per set (Mean)	Vocabulary per set (Min)	Distinct 2 (Mean)	Distinct 2 (Min)	JS Divergence (Mean)	Validation Perplexity (Mean)
Gen 0 (Reference Data)	550.2 ± 6.9	513.0 ± 13.4	0.772 ± 0.016	0.722 ± 0.027	0.334 ± 0.005	32745 ± 11303
s = 0 (Gen 3)	547.1 ± 16.4	499.6 ± 31.2	0.769 ± 0.014	0.693 ± 0.048	0.338 ± 0.008	34996 ± 4853
s = 0.1 (Gen 3)	523.7 ± 25.9	438.8 ± 66.3	0.733 ± 0.040	0.589 ± 0.115	0.342 ± 0.005	37702 ± 4182
s = 0.15 (Gen 3)	501.7 ± 28.4	369.4 ± 136.1	0.703 ± 0.046	0.493 ± 0.219	0.346 ± 0.010	47770 ± 12,034
s = 0.2 (Gen 3)	511.5 ± 26.4	423.2 ± 68.0	0.719 ± 0.042	0.556 ± 0.116	0.343 ± 0.014	45,896 ± 8,807
s = 0.25 (Gen 3)	499.5 ± 20.5	382.6 ± 53.6	0.704 ± 0.031	0.516 ± 0.096	0.345 ± 0.006	60,293 ± 18,033
s = 0.3 (Gen 3)	469.2 ± 69.6	314.0 ± 164.4	0.651 ± 0.113	0.410 ± 0.225	0.370 ± 0.054	53041 ± 8537
s = 0.5 (Gen 3)	341.7 ± 119.2	212.0 ± 208.1	0.464 ± 0.184	0.282 ± 0.292	0.461 ± 0.143	62443 ± 16182
s = 0.7 (Gen 3)	374.9 ± 132.8	256.6 ± 173.4	0.488 ± 0.180	0.321 ± 0.227	0.451 ± 0.186	48469 ± 8733
s = 0.9 (Gen 3)	361.3 ± 96.2	375.6 ± 163.5	0.479 ± 0.142	0.361 ± 0.223	0.415 ± 0.093	64312 ± 18056
Collapse (Gen 3)	381.0 ± 38.8	273.4 ± 95.1	0.502 ± 0.060	0.341 ± 0.130	0.361 ± 0.009	53757 ± 6272

Table 2 : Generation-3 metrics by condition: mean with 95% CI over five seeds

4.3 Repetition Loops (H3)

Repetition loops appeared only at high synthetic shares. 6 of the 20 mixed runs with $s \geq 0.3$ had at least one 1,579-token set that abruptly collapsed into repetition loops despite varied sampling in the previous generation's sample. None of the 25 runs with $s \leq 0.25$ did this.

Each loop phrase occurs in the real text, often only once. For example, "digging, digging, digging" comes from Mr Lorry's dream on the Dover mail, and "a very little indeed" also occurs once in the real text. Mean metrics however hide these failures. At s = 0.5, seed 1337 sample still reported a mean of 271 distinct words per window, even though one window contained only two.

The collapse arm fails differently: it recycles whole memorised passages rather than looping on one phrase. In its worst case (seed 1338, generation 3), one window held only 157 distinct words and an 18% distinct-bigram ratio, and a 20-word passage ("… and with that completeness that if the coach lamps had been blown and stormed out …") was repeated five times. The collapse arm narrows gradually, whereas the mixed-data loops appear abruptly.

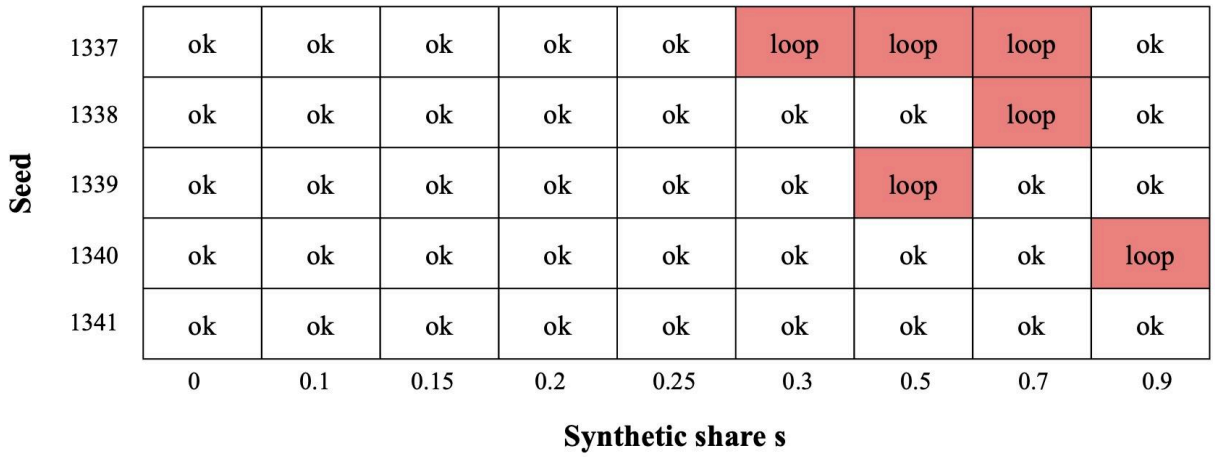


Figure 5 : Repetition loops, by seed and synthetic share. A cell is marked "loop" if at least one set in any generation contains a loop.

s	Seed	Generation	Fewest Distinct Words in a Chunk	Repeated Text
0.3	1337	3	33	"... worse than useless to seek to know ..." (one sentence repeated)
0.5	1337	3	2	"digging , digging , digging ..."
0.5	1339	3	1	"a a a a ..."
0.7	1337	3	2	"you are you are you are ..."
0.7	1338	2	18	"i thank you indeed . i thank you indeed ..."
0.9	1340	3	5	"a very little indeed . a very little indeed ..."

Table 3 : Repetition loops. For each failed run, the table shows the generation and the set with the fewest distinct words, along with the text that was repeated.

4.4 Mean versus Worst-case Metrics

Average diversity metrics do not separate s = 0.25 from s = 0.3. Mean rare-word retention (0.471 and 0.487) and mean vocabulary per set (499.5 and 469.2) are nearly about the same (within one standard deviation).

However, what changed at s = 0.3 was reliability. No run with s = 0.25 showed any breakdown and even its worst set run had 317 distinct words. However, at s = 0.3, one run showed a repetition loop with only 33 distinct words in its worst set.

Mean and worst-case metrics therefore disagree about where the safe limit is, and worst-case checks locate it more precisely.

5. Discussion

The central finding is that real data protects a model against collapse roughly in proportion to its share, but the protection breaks down abruptly, not gradually. Averages describe a gentle decline. Worst-case sets reveal a sharp failure that appears just above the safe limit. Any evaluation that reports only means could conclude that s = 0.3 is safe, when one run in five at that level has already fallen into a loop.

The loops are built from phrases that exist in the real text, so the model is recombining real fragments rather than inventing them. This suggests that synthetic data at high shares does not simply lose information. Instead it re-circulates the same material until it degenerates.

For further works we suggest measuring worst-case sets (repetition, minimum distinct words) alongside means. Third, test the threshold on the actual data and model

6. Limitations

One novel and one small model : Everything here comes from one 13,920-token corpus, a three-layer word-level transformer, and three generations. A heavier-tailed corpus, a larger model, or more generations could move the safe line in either direction.

Perplexity : The absolute perplexity values in this study exceed the uniform-guess ceiling for the vocabulary, so they are not calibrated probabilities. They are used only to compare conditions against each other, never as absolute numbers.

7. Conclusion

Train a model on its predecessor's output, three generations deep, even at this small scale, and the collapse is measurable: two-thirds of the rare vocabulary lost, and the whole vocabulary contracted by 59.6%. Real text stops the decline in proportion to how much of it is in the mix. Between 10% and 25% synthetic, the models held onto 72–88% of the rare tail, with not a single loop among them; at 50% synthetic or more, they ended where pure recursion ends.

The more durable finding concerns how to measure. The means could not tell 25% synthetic from 30%; the worst sets showed a loop at 30% and none at 25%. The safe line is a property of the worst sample, and the average will never surface it. For the scale of this study, the supported operating rule is to keep at least 75% of each generation's data human-written, a synthetic share of 0.25 or less.

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